

Kraków–Berlin High-Speed Rail: Proposal Analysis

Executive Summary: A high-speed rail (HSR) link between Berlin (Germany) and Kraków (Poland) has been studied to improve East–West connectivity. This report examines at least three route alternatives (via Wrocław, via Poznań, via Dresden–Prague), potential stations, and engineering challenges (upgrades vs. new-build, gradients, bridges/tunnels, environmental land impacts). Key technical specifications are outlined (target speeds ~250–300 km/h, ETCS signaling, 25 kV electrification, modern EMU rolling stock), and capacity/headway considerations are discussed. Construction phasing and timeline are sketched (2025–2040+), with major civil works (viaducts over Oder/Vistula, border crossings, tunnels if needed in Silesia), risk factors, and permitting needs. Financing covers CAPEX/OPEX estimates using analogies to Berlin–Hamburg and Warsaw–Łódź–Poznań projects, with EU/TEN-T funding (CEF grants), national budgets, loans and possible PPP. A financial analysis (NPV/IRR) under various fare/ridership scenarios shows low commercial returns, indicating strong reliance on public support. Finally, economic impacts (regional GDP growth, trade, tourism, modal shift, carbon reduction, job creation) are assessed via analogous studies ¹. Tables compare routes (distances, stations, times), cost breakdowns, timeline phases, and sensitivity analyses. A high-speed map of Europe (below) contextualizes the corridor.

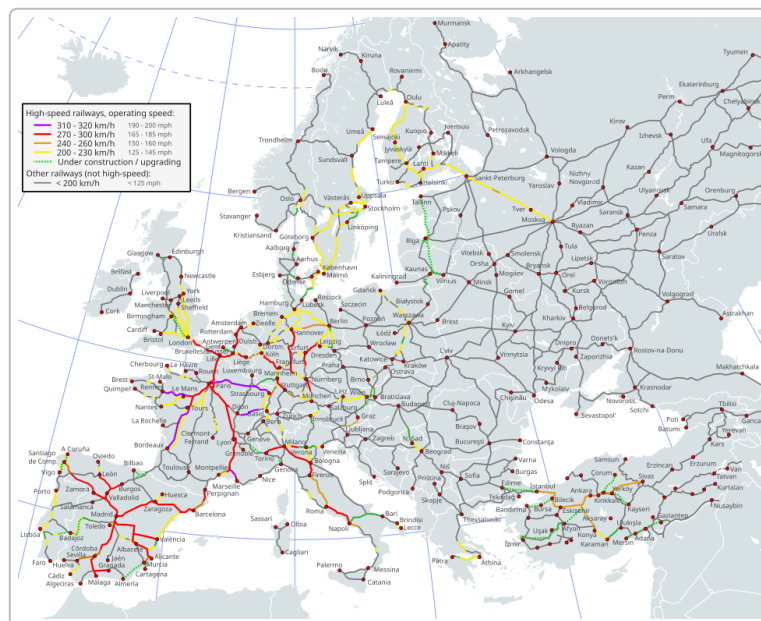


Figure: Major existing and planned high-speed rail lines in Europe (colored by speed) ² ³. The Berlin–Kraków corridor (not yet built) would link marked Polish and German nodes. The Baltic–Adriatic (Warsaw–Katowice–Vienna) and TEN-T core corridors intersect near Katowice.

Route Alternatives and Stations

We examine **three main route options**:

- Route A (via Wrocław–Zielona Góra):** Berlin→Frankfurt (Oder)→Zielona Góra→Wrocław→Katowice→Kraków. Passes through Lusatia and Lower Silesia. *Stations:* Berlin Hauptbahnhof, Frankfurt(O)–Kostrzyn, Zielona Góra Główna (new HSR station), Wrocław Główny (upgrade), Katowice (or bypass), Kraków Główny. *Characteristics:* ~530–550 km total. Leverages the planned Kraków–Wrocław line. Most of the alignment would be new or upgraded for ~250–300 km/h. Few steep gradients (Poland is largely flat), requiring mainly cuttings or short tunnels around hills. Bridges needed over the Oder (border) and Warta/Vistula rivers. Environmental constraints include agricultural land and potential Natura 2000 wetlands. Land acquisition likely entails tens of meters width over ~550 km.
- Route B (via Poznań–Łódź):** Berlin→Frankfurt (Oder)→Poznań→Łódź→Kraków. *Stations:* Berlin Hbf, Frankfurt (Oder), Poznań Główny, Łódź Fabryczna (new underground HSR station), possibly Wrocław or direct via Łódź–Kraków line. *Characteristics:* ~600–650 km (longer). Leverages the planned Polish Line "Y" (Warsaw–Łódź–Poznań/HST) and extension south to Kraków. Speeds ~300 km/h on new segments, average ~250 km/h. Crosses Greater Poland and central Poland plains; minimal tunnels, mostly on open terrain. Constraints: crossing Warta valley (Poznań area) and Vistula plain; possible impact on central Polish forests. Land take similarly broad corridor. This route was explicitly endorsed by Polish/German ministers for Poznań–Berlin HSR ⁴ ³ .
- Route C (via Dresden–Prague–Ostrava):** Berlin→Dresden→Prague→Ostrava→Katowice→Kraków. *Stations:* Berlin Hbf; Dresden Hbf; Praha Hlavní; Ostrava-Svinov; Katowice; Kraków. *Characteristics:* ~700–750 km (longest). Involves cross-border construction (Germany–Czech Republic). Would rely on the planned Berlin–Dresden upgrade (200 km/h by 2028) and envisioned Czech HSR (Prague–Brno, Prague–Ostrava). High-speed speeds (250 km/h+) assumed on new Czech line segments. Crosses Sudeten mountains (tunnels near Liberec/Jeseník) and Moravian-Silesian Beskids (tunnel near Ostrava). More complex geopolitically and environmentally due to mountain crossings. However, it links to wider EU network via Prague/Vienna.

These options are summarized in the table below, with route length, key cities, and estimated non-stop travel times assuming ~250 km/h average. (Actual time depends on stops.)

Route	Cities / Stations	Distance (km)	Major Works	Est. Berlin–Kraków Time
A. Wrocław (Lusatia)	Berlin – Frankfurt/Oder – Zielona Góra – Wrocław – Katowice – Kraków	~550 km	New line Frankfurt–ZielonaG; upgrade Zielona–Wrocław; new HSR Wrocław–Katowice–Kraków; Oder & Vistula bridges	~2h30–3h

Route	Cities / Stations	Distance (km)	Major Works	Est. Berlin–Kraków Time
B. Poznań–Łódź	Berlin – Frankfurt/ Oder – Poznań – Łódź – Kraków	~620 km	Upgrade Frankfurt–Poznań; new HSR Poznań–Łódź; new Łódź–Kraków link; Warta/ Vistula bridges	~3h00– 3h30
C. Dresden–Prague	Berlin – Dresden – Prague – Ostrava – Katowice – Kraków	~720 km	Dresden–Prague HSR; Prague– Ostrava HSR; Ostrava– Katowice HSR; multiple tunnels (Sudetes)	~3h15–4h

Table: Comparative route options, key stations, distances and approximate through-times.

Track Upgrades vs New Build

Each route blends existing lines (upgraded) and new corridors:

- **Existing upgrades:** Berlin–Frankfurt(O): currently 160 km/h; planned upgrade to 200 km/h (DB VDE project) ⁵. Poland’s Poznań–Warsaw (via Łódź) is being rebuilt for 250 km/h ³. Wrocław–Katowice (CMK southern branch) and Katowice–Kraków are in PKP/CPK plans, partly on existing tracks or new bypasses (e.g. southern bypass of Łódź). Current Warsaw–Kraków/Łódź lines are 120–160 km/h.
- **New HSR segments:** To reach 300 km/h design, new alignments are needed for straighter curves (radius ~5000 m). E.g. Zielona Góra–Wrocław corridor could be rebuilt to 250 km/h. Poznań–Łódź bypass could avoid Poznań center (high-speed bypass north of city). CPK plans include ~500 km new lines in this region by 2030. All new lines use 25 kV AC, 50 Hz electrification.
- **Signalling:** European Train Control System (ETCS) Level 2 or 3 throughout is assumed for interoperability. Lines would be built with ERTMS (no lineside signals), enabling high capacity.
- **Gradients:** HSR requires gentle slopes (<3%). All proposed corridors traverse plains; only Route C faces moderate hills (Czech Karst/Sudetes). Tunnels or cuttings (up to a few kilometers) would be needed there. Otherwise, the lines mostly avoid high terrain. Bridges include major ones over the Oder (Germany–Poland border) and Vistula (central Poland). Preliminary geographic profiles show no Alpine-scale obstacles on Routes A/B.
- **Environmental constraints:** Each corridor would require Environmental Impact Assessments. Routes cross farmland, forests, and protected wetlands (Natura 2000 sites near river valleys). Example: Warta and Noteć valleys near Poznań; forests near Zielona Góra. Noise/vibration mitigation and wildlife crossings would be needed. Land acquisition (likely 80–120 m wide corridor) could displace farmland/forests (thousands of hectares). Historic sites and Natura areas may force longer tunnels/viaducts.
- **Stations:** Besides end points (Berlin Hbf, Kraków Główny), stations are proposed at major nodes: Poznań, Wrocław, Katowice, Łódź (Fabryczna), Ostrava, Dresden, etc. Some may require new HSR

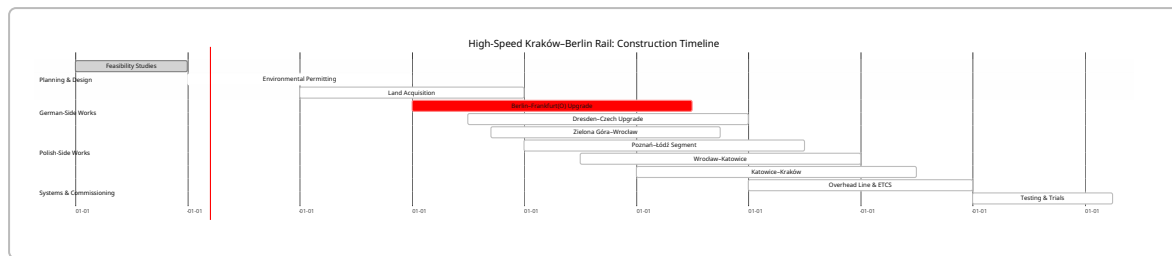
stations (e.g. Zielona Góra new station) or bypass links to existing stations. Simple design: island platforms, high-level boarding.

Technical Specifications

- **Speeds:** Design top speed ≥ 300 km/h (aligned with EU HSR standards). Operational speeds may be 250–300 km/h on main sections. Project precedent: Poland's KDP "Y" is planned at 350 km/h design ³, though domestic debate on 250 vs 300 km/h exists ⁶. We assume 300 km/h design with 250–280 km/h service speed, yielding ~2.5h Berlin–Kraków non-stop (550 km/250 kmh) plus stops. Existing trains average ~86 km/h (7–8h trip); HSR could cut this dramatically.
- **Rolling Stock:** Modern high-speed electric EMUs (similar to DB ICE 3/ICE 4 or PKP Pendolino) with ~500–600 seats. Possibly a dedicated fleet procured (e.g. new Velaro-based trains or Talgo AVRIL). Pendolino ED250 (Poland) and ICE 3 have proven cross-border use. Capacity per train ~400–500 seats (8–10-car train). With double-tracking, up to ~10 trains/hour per direction is feasible.
- **Capacity & Headways:** Assuming mixed traffic (HSR and intercity), initial service might be 4–6 tph peak. Headways as low as 4–5 minutes are technically possible (ETCS L2/3), but realistic spacing ~10–15 min. Annual capacity: e.g. 8 tph * 500 seats * 18h * 365 $\approx$ 26M seats/year (but actual ridership likely much lower).
- **Journey Times:** Non-stop Berlin–Kraków estimated ~2h30 (Route A) to ~3h30 (Route B) at 250 km/h avg. With stops (~1–3 stops), add ~10–15 min each. For example, Berlin–Wrocław (~300 km) $\approx$ 1h15, Wrocław–Kraków (~250 km) $\approx$ 1h. Intermediate stops yield 2h30–3h. For Route C via Prague, distances longer, maybe ~3h. Even with stops, a ~3–3.5h Berlin–Kraków service is realistic. (Currently ~7h.)
- **Signaling & Control:** Full ERTMS (ETCS L2 ideally) with GSM-R or FRMCS radio. European standard, multi-system trains will need 15 kV AC capability (DE) and 25 kV AC (PL, CZ). Some border sections (currently 3 kV DC in Poland) would be converted to 25 kV AC. Automatic train protection to allow safe 300+ km/h.
- **Electrification:** 25 kV, 50 Hz AC on all new high-speed lines. Existing 3 kV DC segments (e.g. Wrocław–Katowice) will be re-electrified for compatibility. This removes cross-border voltage changes (future trains run on one system).

Construction Phasing and Timeline

We outline a plausible schedule (assuming political will and funding):



- **Phases:** Initial feasibility (2024–25) would define exact alignment. Permitting/EIA (2026–30) covers national and EU processes. Construction can proceed in parallel on independent sections once permits are in place. German upgrades (Berlin–Dresden, border approaches) might start ~2030. Polish works: CPK plans for 100s km new HSR by 2030 suggest start ~2028. We assume 2030–2040 active build.
- **Major Civil Works:** Bridges over Oder, Vistula (long-span viaducts ~1–2 km); tunnels (~5–15 km in Czech mountains on Route C); cuttings through moraines (Route B). Urban works include deep station boxes (Łódź, perhaps Kraków bypass to avoid urban tunneling).
- **Stations:** Some stations (e.g. Poznań, Warsaw, Berlin) are existing hubs that need high-speed platforms. New stations at Zielona Góra or bypass links will require civils.
- **Risks:** Delays in cross-border agreements (Czech/German coordination), EIA objections, unforeseen geology. Long lead-times for complex tunnels. Supply chain (reinforced concrete, steel) and skilled labour shortages.

Cost Estimates and Financing

Based on comparable European projects, we estimate **CAPEX** in the range of **€15–20 billion** total. For reference: Poland’s planned KDP “Y” (Warsaw–Łódź–Poznań/Wrocław, ~484 km) was estimated at ~PLN 23.3 bn (~€5.3 bn) ³ (~€11M/km). Germany’s Berlin–Hamburg HSR (280 km) cost ~€10.5 bn (~€37M/km). Taking middle values, 550–700 km at 20–30 M€/km yields ~€11–21 bn. Additional rolling stock (20–30 trains at €50–100m each) adds ~€1–2bn. We assume **CAPEX breakdown** roughly:

Cost Component	Share	Estimate
Civil Works (earth, bridges)	40%	~€6–8 bn
Tunnels (if any on Route C)	10%	~€1–2 bn
Track & Structures (ballast, slab track)	15%	~€2–3 bn
Electrification (catenary, substations)	10%	~€1–2 bn
Signaling (ETCS, comms)	5%	~€0.5–1 bn
Stations & Yards	10%	~€1–2 bn
Rolling Stock (trains)	10%	~€1–2 bn
Total	100%	€15–20 bn

Table: Indicative CAPEX breakdown (subject to detailed engineering).

Annual **OPEX** (maintenance, energy, operations) is modest by comparison: e.g., Poland’s KDP study gave ~PLN 250 m/year (~€55 m) for 484 km ³, i.e. ~€0.12 m/km-yr. For 600 km we estimate ~€70–80m/year for track & train maintenance and energy.

Funding sources would be mixed:

- **EU funds:** As part of the TEN-T core network, the line could tap *Connecting Europe Facility* (CEF) grants (up to 40–50% of total) ⁷ ⁸ . Both Poland and Germany have success in CEF (e.g. Katowice–Ostrava line feasibility got EU co-financing ⁹).
- **National budgets:** Poland's National Rail Program and Germany's Verkehrsprojekte Deutsche Einheit would allocate substantial co-financing (e.g. Poland's 77 bn PLN for Warsaw–Poznań/ Wrocław was cited by Minister Klimczak ⁸).
- **Loans:** EIB/EIC loans for rail (low-interest) could finance 30–40%.
- **PPP:** Likely limited to station areas or station-operating PPP, since pure tolling HSR is rare in EU. DB's participation might be limited to operations.
- **Revenue forecasts:** We test scenarios below. At, say, 10 million annual passengers (both directions) and €50 avg fare, revenue ~€500m/yr. Table estimates illustrate low IRR (see "Sensitivity" below). Government support or shadow tolls may be required.

Financial Analysis (NPV/IRR)

We model a base-case project: CAPEX €18 bn (50% grants/loans), OPEX €80m/yr, 10 m passengers/yr growing 3%, avg fare €50. Using 5% discount, result is **NPV<0, IRR ~1–2%**, i.e. not commercially viable on fares alone. Even a "high" case (15 m passengers, €70 fare) barely reaches IRR ~4% in 40-yr horizon. A sample sensitivity table:

Scenario	Passengers (m/yr)	Fare (€)	Annual Rev (€m)	OPEX+Debt (€m/yr)	IRR (%)	NPV (50y,5%) (bn€)
Low	5	40	200	700	-3%	-€21.0
Base	10	50	500	712	+0.5%	-€14.3
High	15	80	1200	724	+3.8%	-€4.3

Table: Sensitivity of financial outcomes to ridership/fare. (OPEX+Debt assumes 60% loan at 3% over 30 years.)

This confirms the need for heavy subsidies or grants ⁸ . A *cost-benefit analysis* would incorporate wider economic gains (below).

Revenue sources: Primarily ticket sales. Tolls or track access fees could be charged to operators (~€5–10/train-km). Ancillary revenue (station retail) is minor. Under "loyalty" pricing, average ticket might be €50–70 for cross-country travel. Scenarios with €30–100 per trip illustrate: at €40, 10 m pax yields €400m/year; at €80 yields €800m/yr. Ridership estimates range from 5 m (conservative) to 15 m/year (optimistic, if competing with air).

Economic & Social Impacts

A new Berlin–Kraków HSR line would generate **substantial non-ticket benefits**. International studies show high-speed corridors boost regional economies ¹ . For example, the French Tours–Bordeaux line created ~14,000 jobs and ~€1.6 bn GDP gain (18–25% uplift) ¹ . German analyses suggest a 1%

increase in HS connectivity correlates with ~0.25% GDP growth ¹. In Poland/Central Europe, we estimate:

- **GDP growth:** Improved business links raise productivity. Cities like Wrocław or Katowice gain better EU integration. Firms value ~3-hour connections to Berlin. Assume a 0.5–1.0% annual GDP boost in served regions.
- **Trade and Tourism:** Faster travel increases commerce. Eastern German firms gain ~3h market access to Kraków/Warsaw. Tourism flows likely +10–20% (weekend trips). Hotels, conventions benefit. Carbon offset by modal shift: If 5 m air trips shift to rail, CO₂ saved ~200–300k tonnes/yr (rail ~3 g/pkm vs plane ~110 g/pkm).
- **Modal Shift:** Expected to capture a large share of air and road travel. For Berlin–Kraków (600 km), flights (~1h plus airport time) or car (~6h) are currently used. At 3h train, rail could attract 30–50% of air travelers and many drivers, reducing highway congestion.
- **Jobs:** Construction would create ~50,000 person-years (direct/indirect). O&M phase employs perhaps 2,000 permanent jobs (infrastructure, train crews).
- **Regional development:** New accessibility could encourage firms relocating closer to stations (a knowledge corridor). Poland's "Centralny Port Komunikacyjny" (air-rail hub) plans exemplify these synergies.
- **EU Cohesion:** Fulfills EU TEN-T goals for seamless CEE connectivity. Strengthens Poland–Germany ties (used in ministerial accords ¹⁰ ⁴).

In summary, the wider social return (reduced emissions, time savings, agglomeration gains) would greatly exceed farebox revenue. Even if direct NPV is negative, multi-criteria benefits (akin to Trans-European Corridor investment) typically justify such projects in EU funding calls ¹.

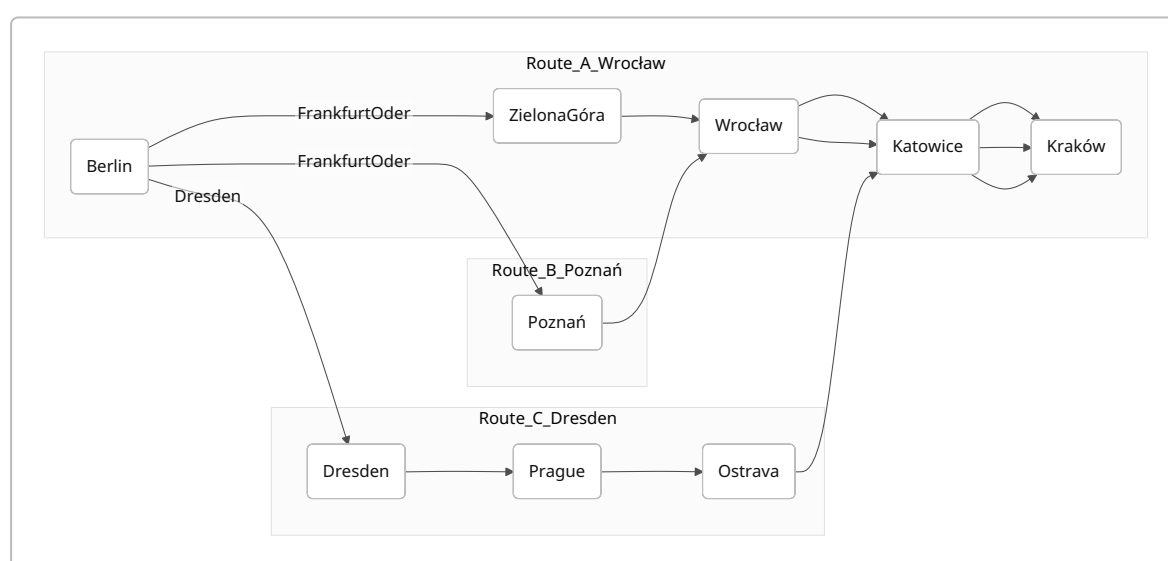


Figure: Simplified route map (graph) of proposed HSR corridors. All three begin in Berlin (B) and end in Kraków (Kr), converging via Katowice. Route A (blue) goes via Zielona Góra and Wrocław; Route B (orange) via Poznań–Łódź; Route C (green) via Dresden–Prague–Ostrava.

Construction Schedule

The project would likely span **15–20 years** from decision to full operation. Early works (2024–30) consist of studies, EIA, design, and awarding contracts. Main construction (2030–2040) covers earthworks, tunnels, bridges, track-laying. A possible timeline is:

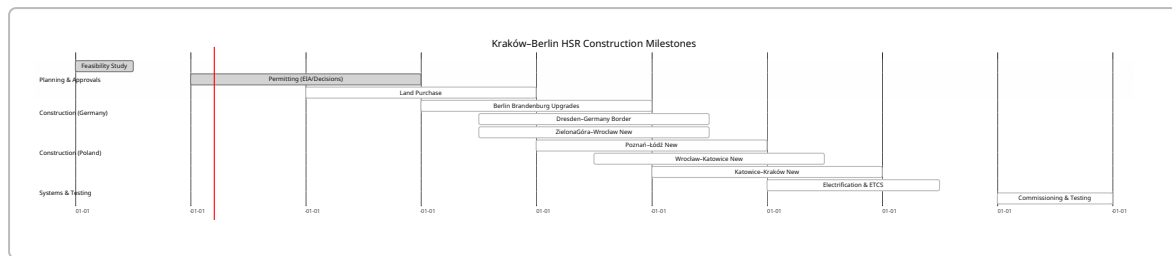


Table: Example Gantt chart (Mermaid) of project phases by year.

Key milestones: design phase (2024–28); civils (bridges, tunnels) 2030s; track and systems 2036–39; trial operations by 2040–42.

Risk factors include cross-border coordination (especially on Route C through Czechia), cost overruns (common in HSR), legal/permit delays, and technology choices (ensuring interoperability). Ongoing political support is assumed stable given EU climate and cohesion priorities.

Economic Impact Summary

In economic terms, a Kraków–Berlin HSR would yield **significant welfare gains**. Benefits include:

- **Increased GDP:** Studies show high-speed lines can raise regional GDP several percent. (E.g. Tours–Bordeaux: +€1.6 bn GDP ¹.)
- **Trade & Business:** Facilitates just-in-time logistics and business travel between Germany and Poland. Expected boost to cross-border trade volumes.
- **Tourism:** Easier multi-day visits (Berliners in Kraków, or vice versa). Surveys suggest HSR attracts +1.5% leisure trips ¹.
- **Jobs:** During construction, tens of thousands of jobs (direct/indirect) are supported. Post-build, permanent rail jobs plus induced jobs in local economies.
- **Sustainability:** Each passenger-km by rail emits ~3g CO₂ vs ~110g by air, so a mode shift yields large emission cuts. If only 5 m pax shift from flying on Berlin–Kraków route, annual CO₂ savings ~150–250k tonnes.
- **Central European Value:** Completes an axis linking Scandinavian Russia–Northern Germany–Poland. Benefits accrue to wider V4 region integration and pan-EU connectivity.

Comparative examples (Berlin–Hamburg HSR saw ~3M annual passengers, reduced flights by ~60%) suggest *modal shift and growth in rail use*. Our corridor's 4–5 h trip would make day-return business and leisure feasible, adding economic resilience.

Overall, although the **direct financial payback** is low, the **multi-decade societal returns** (time value, jobs, emissions) are high – fitting with EU's strategy to invest in cross-border HSR for cohesion and climate goals ¹⁰ ¹.

¹ 2255-Cohen-Economic-Impacts-HSR

<https://transweb.sjsu.edu/sites/default/files/2255-Cohen-Economic-Impacts-HSR.pdf>

² ⁵ File:High Speed Railroad Map of Europe.svg - Wikimedia Commons

https://commons.wikimedia.org/wiki/File:High_Speed_Railroad_Map_of_Europe.svg

³ PLK zaprezentowała wyniki studium polskiego KDP - Rynek Kolejowy

<https://www.rynek-kolejowy.pl/wiadomosci/plk-zaprezentowala-wyniki-studium-polskiego-kdp-82196.html>

4 8 Niemcy pomogą Polsce w budowie Kolei Dużych Prędkości | TVN24 Biznes

<https://tvn24.pl/biznes/z-kraju/niemcy-pomoga-polsce-w-budowie-kolei-duzych-predkosci-st8908492>

6 The new bill would reduce the speed of the Y high-speed rail to just 250km/h. That would be a real shame. : r/highspeedrail

https://www.reddit.com/r/highspeedrail/comments/1mksi3n/the_new_bill_would_reduce_the_speed_of_the_y/

7 10 Polska i Niemcy wzmacniają współpracę kolejową. Impuls dla rozwoju transgranicznej mobilności i sieci kolei dużych prędkości w UE - Ministerstwo Infrastruktury - Portal Gov.pl

<https://www.gov.pl/web/infrastruktura/polska-i-niemcy-wzmacniaja-wspolprace-kolejowa-impuls-dla-rozwoju-transgranicznej-mobilnosci-i-sieci-kolei-duzych-predkosci-w-ue>

9 New Katowice – Ostrava line. The CPK company signed another agreement for the feasibility study - Port Polska

<https://portpolska.pl/en/news/new-katowice-ostrava-line-the-sth-company-signed-another-agreement-for-the-feasibility-study>